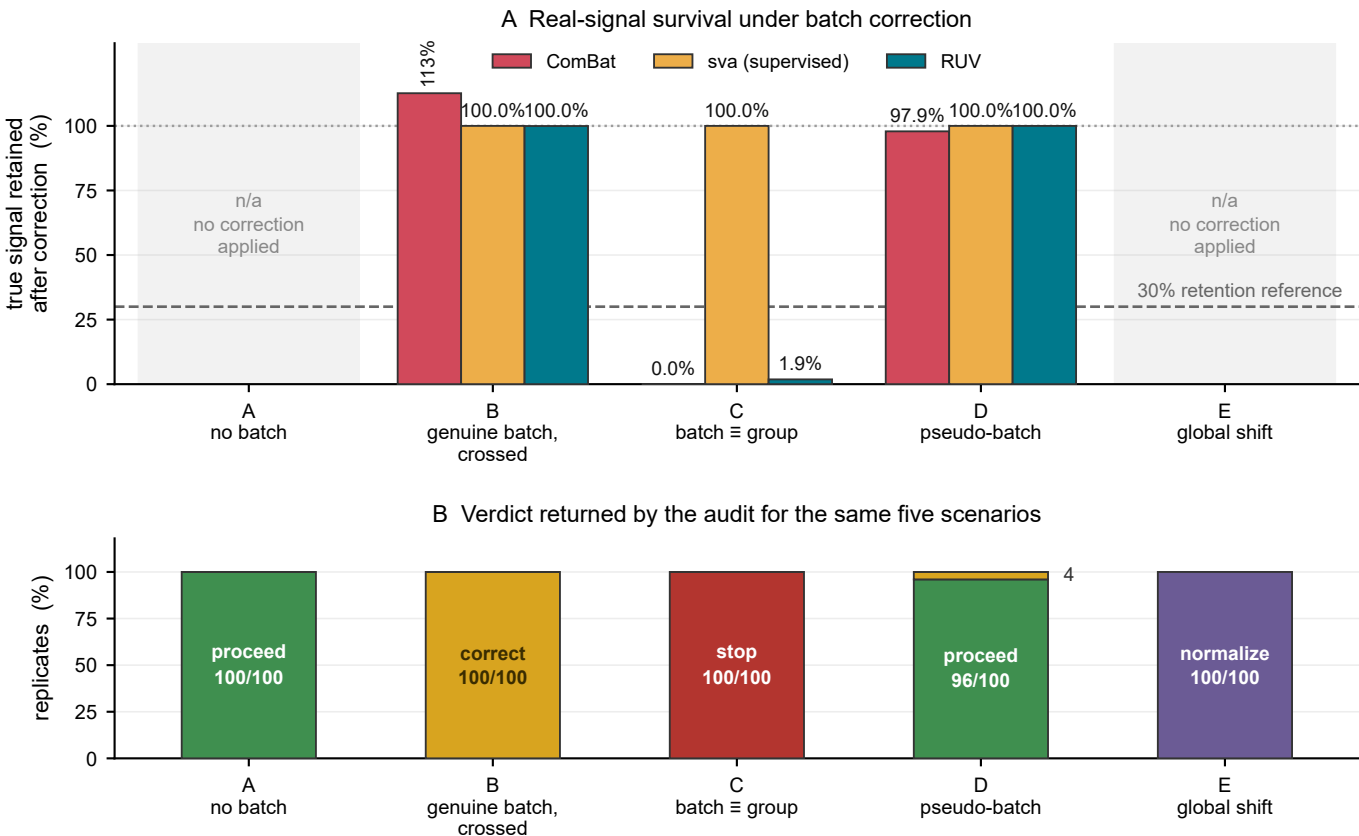


Fig. 3 Correction tools erase real signal under confounding and say nothing



100 replicates per scenario, 2,000 genes, $n = 15$ per group, 200 permutations. sva is run in supervised mode (group labels supplied). In B, ComBat returns $>100\%$ because it inflates rather than preserves the contrast. None of the three tools has any design-admissibility output, so none of them can report scenario C as unfit; only the audit refuses it.